

Metric Thresholds & Operating Ranges

Future Grid-Stress Scenario Explorer | August 5, 2026 | Winter Analysis Only

This document resolves Blocker 2 by defining the acceptable operating range, warning range, and failure threshold for each of the four metrics that determine whether a given year earns a point in the Scenario Robustness Score. All derivations use winter data only (December-February) and V1.0 metric formulas. Summer thresholds are deferred.

A year earns its Robustness Score point only when all four metrics simultaneously meet their Acceptable threshold. The maximum possible score is 5 - one point per analysis year (2022-2026).

Implementation note for Mitchell: All four metrics should be computed per identified stress event, not across the entire winter dataset. W06 is one event, W03 is another. Each earns or fails its year's robustness point independently. For years with multiple events (e.g., 2025: W02, W03, W04), the year passes if any single event meets all four thresholds.

Winter Data Anchors

Anchor Value	Winter (Dec-Feb)
Stressed hours p90 threshold	18,413 MWh/hr
Total stressed hours (2022-2026)	108 hours
Avg load during stressed hours	~19,000-19,500 MWh/hr
Capacity deficit per stressed hour	avg load minus threshold = ~600-1,100 MWh/hr above p90
Total winter capacity deficit (est.)	600 MWh/hr avg x 108 hours = 64,800 MWh
W06 capacity deficit (est.)	800 MWh/hr avg x 54 stressed hours = 43,200 MWh
W06 fossil generation (stressed hours)	Oil 4,000 MW/hr + Gas ~7,920 MWh/hr x 54 hours = 643,680 MWh
W06 duration / stressed hours	18 days / 54 stressed hours (3 hrs/day x 18 days)
Avg winter wind (stressed hours)	756.9 MWh/hr
Protected Reserve	30% of Total Capacity
Max Available Charge	70% of Total Capacity

Metric 1 - Capacity Margin Deficit Reduction (CMDR)

Corrected Formula

$$\begin{aligned}\text{CM}(t) &= \max(0, \text{CM_with_storage}(t) - \text{CM_without_storage}(t)) \\ &= \max(0, \text{capacity_dispatched}(t))\end{aligned}$$

CM_deficit_without_storage(t) is the capacity shortfall at each stressed hour without storage: RTLO(t) minus the p90 threshold (18,413 MWh/hr). CMDR% measures what fraction of the total winter capacity deficit storage eliminated. The max(0,...) floors any negative contribution at zero. Score maps linearly: 0 at CMDR = 0%, 100 at CMDR = 100%.

Derivation at Pilot-Scale Dispatch Levels (W06 anchor)

W06 capacity deficit = 800 MWh/hr avg excess x 54 stressed hours = 43,200 MWh

At 75 MW dispatch x 54 hours = 4,050 MWh -> CMDR% = 9.4% -> Score 9
At 200 MW dispatch x 54 hours = 10,800 MWh -> CMDR% = 25.0% -> Score 25
At 500 MW dispatch x 54 hours = 27,000 MWh -> CMDR% = 62.5% -> Score 63
At 921 MW dispatch x 54 hours = 49,734 MWh -> CMDR% = 115% -> Score 100
(capped)

Operating Ranges

Zone	Threshold	Score	Meaning
Acceptable	CMDR >= 20%	>=20	Storage eliminates at least 20% of winter capacity deficit during stressed hours
Warning	5% to 19%	5-19	Measurable deficit reduction but below minimum meaningful threshold
Failure	<= 0%	0	No capacity deficit reduction - year fails robustness point

Robustness pass condition: CMDR% >= 20% (Score >= 20). Achievable at ~200 MW avg dispatch during W06-length events.

Metric 2 - Stress Window Effectiveness (SWE)

Formula

SWE = Sum(capacity_dispatched(t)) / Sum(oil_actual(t) + gas_actual(t))
Computed over stressed hours of each identified event only

Denominator is fossil fuel generation only - oil plus gas. Wind is excluded. This makes SWE a direct fossil displacement metric. Score maps linearly: 0 at SWE = 0%, 100 at SWE = 100%.

Derivation (W06 stressed hours anchor)

W06 fossil during stressed hours:
Oil: 4,000 MW/hr x 54 hours = 216,000 MWh
Gas: ~7,920 MWh/hr x 54 hours = 427,680 MWh
Total fossil stressed window = 643,680 MWh

At 200 MW x 54 hours = 10,800 MWh -> SWE = 1.7% -> Score 2
At 500 MW x 54 hours = 27,000 MWh -> SWE = 4.2% -> Score 4
At 921 MW x 54 hours = 49,734 MWh -> SWE = 7.7% -> Score 8

Across all 108 winter stressed hours (not just W06):
Total winter fossil ~ 1,287,360 MWh
At 500 MW x 108 hours = 54,000 MWh -> SWE = 4.2% -> Score 4

Operating Ranges

Zone	Threshold	Score	Meaning
Acceptable	SWE >= 3.0%	>=3	Storage displaces at least 3% of fossil generation during winter stressed hours
Warning	0.5% to 2.9%	1-2	Measurable displacement but below meaningful threshold at system scale
Failure	< 0.5%	0	Negligible fossil displacement - year fails robustness point

Robustness pass condition: SWE >= 3.0% (Score >= 3). Achievable at ~500 MW avg dispatch during W06-length events.

Metric 3 - Fuel Offset Percent (FOP)

Formula

```
FFGO = Sum(oil(t)) + Sum(gas(t)) - (Sum(wind(t)) +
Sum(capacity_dispatched(t)))
FOP = (Sum(wind(t)) + Sum(capacity_dispatched(t))) /
Sum(total_generation(t)) x 100
```

FOP measures the combined share of total generation covered by wind plus storage during the event. Score maps linearly: 0 at FOP = 0%, 100 at FOP = 100%. Wind alone contributes ~4.1% of NE system winter generation during stressed hours - storage adds incremental offset on top.

Derivation (all 108 winter stressed hours)

```
Total winter generation across 108 stressed hours:
~18,500 MWh/hr x 108 hours = 1,998,000 MWh

Wind contribution (756.9 MWh/hr x 108 hrs) = 81,745 MWh
Wind FOP baseline = 81,745 / 1,998,000 x 100 = 4.1%

Adding storage:
At 200 MW x 108 hrs = 21,600 MWh -> FOP = (81,745+21,600)/1,998,000 = 5.2% ->
Score 5
At 500 MW x 108 hrs = 54,000 MWh -> FOP = (81,745+54,000)/1,998,000 = 6.8% ->
Score 7
At 921 MW x 108 hrs = 99,468 MWh -> FOP = (81,745+99,468)/1,998,000 = 9.1% ->
Score 9
```

Operating Ranges

Zone	Threshold	Score	Meaning
Acceptable	FOP >= 5.0%	>=5	Wind + storage jointly offset at least 5% of total winter generation during the event
Warning	2.0% to 4.9%	2-4	Combined contribution present but marginal at NE system scale
Failure	< 2.0%	0-1	Wind + storage negligible relative to total generation - year fails robustness point

Robustness pass condition: FOP >= 5.0% (Score >= 5). Achievable at ~200 MW avg storage dispatch combined with average winter wind.

Metric 4 - Recharge Capacity Mismatch (RCM)

Formula

```
Cycle_Recharge_Mismatch(t) = recharge_opportunity(t) - actual_recharged(t)
Avg_Recharge_Mismatch      = (1/N) x Sum[cycle_recharge_mismatch(t)]
RCM                        = avg_recharge_mismatch /
maximum_available_capacity
```

Maximum available capacity = 70% of Total Capacity (per project constraints). RCM = 0% is perfect balance. Positive RCM means undercharging - available wind not being captured fast enough, battery depletes progressively. Negative RCM means overcharging opportunity - battery too small for available wind, wind being wasted. Both extremes reduce the score. Scoring table is already defined in project spec V1.0.

Winter Derivation

```
Max available capacity      = 70% of Total Capacity
Winter avg wind off-peak    = 756.9 MWh/hr x 21 off-peak hours = 15,895 MWh/day
Protected reserve floor     = 20% of Total Capacity (cannot discharge below)

RCM = 0%: Perfect balance - no wasted wind, no depleted battery
RCM = +5%: Undercharging - battery absorbing 5% less than max available/cycle
          Battery depletes progressively across W06-length events
RCM = -5%: Overcharging opportunity - 5% more wind available than battery holds
          Wind being curtailed or wasted
```

Operating Ranges

Zone	Threshold	Score	Meaning
Acceptable	0% to +5.0%	75-100	Recharge closely matches winter wind opportunity - system in balance across multi-day events
Warning	+5.0% to +10.0%	35-74	Meaningful mismatch - battery risks depletion before W06-length events conclude
Failure	> +10.0%	0-34	Severe mismatch - storage strategy cannot sustain multi-day winter events - year fails robustness point

Robustness pass condition: RCM Score >= 75 (mismatch <= +5.0%). Full RCM scoring table defined in project spec V1.0.

Summary - All Four Winter Thresholds

All four metrics must simultaneously meet their Acceptable threshold for a year to earn its Robustness Score point. Partial credit is not awarded.

Metric	Acceptable	Warning	Failure	Pass Condition
CMDR	>= 20% (Score >= 20)	5% to 19%	<= 0%	CMDR% >= 20%
SWE	>= 3.0% (Score >= 3)	0.5% to 2.9%	< 0.5%	SWE >= 3.0%
FOP	>= 5.0% (Score >= 5)	2.0% to 4.9%	< 2.0%	FOP >= 5.0%
RCM	0% to +-5% (Score >= 75)	+5% to +-10%	> +-10%	RCM Score >= 75

Derivation basis: All thresholds calibrated against NE system-wide winter stressed hour data (p90 = 18,413 MWh/hr, 108 total stressed hours across 2022-2026) anchored on W06 fossil fuel generation from ISO-NE Winter 2026 operational report (oil avg 4,000 MW/hr; gas 44% of supply mix across 18 consecutive days). Thresholds represent minimum meaningful contribution at NE system scale for a pilot-scale seafloor storage deployment. Summer thresholds are deferred and will be addressed separately.

